Spacetime Algebra as a Theorem: Deriving Cl(3,1) from the Structure of a Dynamical Unit

Henry Molina
Independent researcher henrymolina@gmail.com DOI: 10.5281/zenodo.21184515

Self-contained manuscript; requires no external framework beyond standard linear algebra and Clifford algebra conventions. Numerical verifications referenced in §7 are at:
https://github.com/hmolinab/papers/tree/main/weld_clifford/code

---

Abstract

We prove that any operative dynamical unit (ODU) is necessarily an element of the real Clifford algebra $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$, given four structural axioms, the fourth of which makes explicit a co-location premise otherwise left implicit. A1 (SAIR): the unit is described by four intrinsic attributes: a scalar S and three vectors $\mathrm{A}, \mathrm{I}, \mathrm{R}$ in $\mathbb{R}^d$. A2 (geometric product): structure is governed by the geometric product of those attributes, whose grade-2 part $\mathrm{I} \wedge \mathrm{R}$ is the Field bivector. A3 (continuous evolution): the ODU evolves smoothly in time and space at finite propagation speed, with the attribute space and the propagation coordinates identified (A3′, co-location). From these four axioms, without postulating a spacetime metric or background geometry, we derive: (i) the closure condition $\binom{d}{2} = d$ forces $d = 3$ uniquely (Hodge self-duality of bivectors in $\mathbb{R}^3$, confirmed by Hurwitz); (ii) smooth evolution (A3) requires a fourth temporal direction independent of the spatial attributes; (iii) the principal symbol of the resulting second-order PDE is the Minkowski form $\eta = \mathrm{diag}(-1, +1, +1, +1)$, whose real Clifford algebra is $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$. Three closing propositions establish that orthonormality is gauge-redundant (P1), that γ_0 is the algebraic generator conjugate to ∂_τ in the Dirac factorization of $\Box$ (P2, without conflating algebra elements with differential operators), and that the Frobenius norm is the unique Clifford inner product forced by A2 (P3). Within the SAIR framework, Cl(3,1) is a derived consequence rather than a geometric postulate. Classical mechanics and free electrodynamics appear as structural limits. The theorem is conditional: it presupposes that a domain admits $S, \mathrm{A}, \mathrm{I}, \mathrm{R}$ as grade-0/grade-1 attributes in the first place, a premise this paper does not establish for any specific domain (§8.4).

Keywords: Clifford algebra, geometric algebra, Hurwitz theorem, spacetime signature, dynamical systems, matrix normal form, Frobenius metric.

---

1. Introduction

The Clifford algebra $\mathrm{Cl}_{3,1}$, equivalently the spacetime algebra (STA) of Hestenes (1966), is the standard algebraic scaffolding for special relativity and Dirac theory. Its standard motivation is geometric: one postulates a Minkowski spacetime with signature $(3, 1)$ and then constructs the associated Clifford algebra. The question we address is different: is the Lorentzian signature a theorem, rather than a postulate, if one asks what algebraic structure a self-describing dynamical unit must have?

We show that the answer is yes, under four minimal axioms: A1 (attribute structure), A2 (geometric product), A3 (smooth evolution at finite speed), and A3′ (co-location of the attribute space with the propagation coordinates). The derivation does not require spacetime as an input; the signature emerges from the principal symbol of the equation of motion dictated by A3.

This paper is part of a larger program, the Gamma Space Framework (GSF; Molina 2025), whose central object is a real 4×4 configuration matrix $\Gamma \in M_4(\mathbb{R})$. The present paper establishes the algebraic foundation: that Γ is an element of $\mathrm{Cl}_{3,1}$, not by postulate but by necessity. The companion paper (Molina 2024a) establishes the dynamical result: that the determinant of Γ is the source of the cubic term in the soft-mode reduction of the matrix gradient flow. The term operative dynamical unit (ODU) used throughout is self-contained and requires no prior knowledge of the GSF.

Relation to the geometric algebra literature. The spacetime algebra program (Hestenes 1966, 1986; Doran and Lasenby 2003) is the closest antecedent. That program takes Minkowski spacetime as given and develops physics in terms of $\mathrm{Cl}_{1,3}$ (one time, three space, in Hestenes' convention). The signature choice is not merely a convention: $\mathrm{Cl}_{1,3} \cong M_2(\mathbb{H})$ (quaternionic), whereas $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$ (real). These are non-isomorphic as real algebras. The present derivation forces $\mathrm{Cl}_{3,1}$, not $\mathrm{Cl}_{1,3}$, because we require Γ to be a real matrix (dissipation and gradient flows are real processes); this distinguishes the two conventions at the algebraic level. The derivation does not compete with the spacetime algebra program; it identifies which real algebra is forced by the structure of any evolving dynamical unit, and provides a structural basis for that program.

Plan. §2 states the four axioms. §3 derives the four lemmas. §4 states and proves the main theorem. §5 establishes the three closing propositions. §6 illustrates with two physical limits (Newton and Maxwell). §7 gives numerical verification of key steps. §8 discusses scope, related work, and open problems.

---

2. Axioms

We consider an operative dynamical unit (ODU): an entity that (i) exists as a coherent whole distinct from its environment, (ii) acts upon its environment, (iii) has an intrinsic drive, and (iv) is embedded in a relational context. No further phenomenological content is assumed.

Axiom A1 (SAIR attribute structure). Any ODU is completely described at the structural level by four intrinsic attributes: - $S \in \mathbb{R}$ (Singularity, grade 0; identity / self-measure) - $\mathrm{A}, \mathrm{I}, \mathrm{R} \in \mathbb{R}^d$ (Agency, Impulse, Relation; grade-1 vectors)

where d is to be determined. The mapping from observable properties of any coherent entity to the four structural slots $\{S, \mathrm{A}, \mathrm{I}, \mathrm{R}\}$ is claimed to be structurally unique (no two inequivalent assignments produce structurally identical predictions for the same entity), but this paper establishes that claim only at the level of the container, not of instances, and the two should not be conflated (see the qualification at the end of Remark 2.1).

Remark 2.1. A1 is the foundational axiom of the framework; it is not derived from simpler premises within this paper. Its justification is the structural argument that $\{S, \mathrm{A}, \mathrm{I}, \mathrm{R}\}$ are the grades of a geometric algebra of minimal dimension consistent with A2, a circularity resolved by the mutual consistency of A1 and A2, not by an independent proof of A1. The role of A1 in this structure is analogous to that of natural selection in Darwinian theory: a minimal posit that generates the rest.

Qualification (what "structurally unique" means here, and what it does not). A1's uniqueness clause is proved in this paper only for the container: given that an ODU has one grade-0 and three grade-1 attributes at all, Lemmas 2–4 show the algebra hosting them is forced to $\mathrm{Cl}_{3,1}$, uniquely, by a Schur-type argument (representation compatibility) that is exactly why the grades cannot be reshuffled once fixed. This paper does not establish uniqueness at the level of instances: given a specific ODU (a particle, a cell, a market), which observable quantity fills S versus A versus I versus R is an assignment problem the container theorem is silent on. A necessary condition for that assignment (candidates must share the slot's representation under the domain's covariance group) is the same Schur argument specialized to instances, and a companion line of work develops sufficient selection criteria and tests them against seven worked domains, but that work is at an earlier stage of rigor than this paper's closed lemmas and is deliberately not imported here (see §8.4). The word "unique" in A1 should be read as "unique at the container level, proved; open at the instance level" until that companion work matures. This does not weaken A1; it applies the same discipline as §8.4: state exactly what is proved, and do not let a strong word in an axiom imply more than the theorem delivers.

A second, prior qualification: existence, not just uniqueness. Everything above concerns uniqueness of the slot assignment given that a well-posed SAIR quadruple already exists for a domain. A separate and more basic question is existence: does a given domain admit A, I, R as grade-1 vectors at all, with Force and Field genuinely inherent to it? The theorem of this paper is conditional on a positive answer, and does not itself supply one. Existence requires that the domain itself satisfy A1's own preconditions, a coherent whole with a genuine spatial/relational embedding, not merely that some observable be relabeled as a vector; establishing it for a specific domain is an empirical question outside the scope of this paper, which proves only the container theorem — that if the four attributes exist, their host algebra is forced. Read correctly, the Main Theorem (§4) says: "if a domain has grade-1 A, I, R, then their host algebra is forced to $\mathrm{Cl}_{3,1}$," not "every domain has such attributes." The container theorem is proved unconditionally as a piece of algebra; its applicability to a specific domain is not, and should never be read off this paper alone.

An ODU is not merely a system with four labelable slots. The attributes are intrinsic in the operational sense: they are the generators of the geometric product of A2, not observational labels assigned from outside. A threshold relay (e.g., a thermostat) has four assignable properties but satisfies neither A2 (its dynamics are a discontinuous switching rule, not a geometric product) nor A3 (its evolution is not smooth). The word "intrinsic" in A1 is therefore grounded by A2 and A3 together, not declared by fiat.

Axiom A2 (geometric product dynamics). The dynamics of an ODU is governed by the geometric product of its attributes. In a geometric algebra $G(d)$ over $\mathbb{R}^d$, the geometric product of two grade-1 elements u, v decomposes canonically:

$$uv = u \cdot v + u \wedge v$$

into a symmetric (grade-0) scalar part and an antisymmetric (grade-2) bivector part. Applied to the attributes: the Force sector Γ_s is the symmetric Gram coupling of the scalar identity S with the grade-1 attributes $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$, encoding the metric structure of the unit. The Field $\mathcal{F} = \mathrm{I} \wedge \mathrm{R} \in \Lambda^2(\mathbb{R}^d)$ (grade-2, antisymmetric) is the reactive sector. "Force" names the symmetric sector of Γ; S is grade-0 and $S\mathrm{A}$ is a grade-1 vector, not a scalar; the symmetric structure enters through the Gram matrix, not through a grade-0 product. The Force/Field split is algebraically forced by A2, not a separate postulate.

Definition (SAIR embedding: the matrix construction named, gauge closed). The phrase "Gram coupling" above names an operation without writing it; we write it once, explicitly. Embed the vector attributes as the columns of $W = [\,\mathrm{A} \mid \mathrm{I} \mid \mathrm{R}\,] \in \mathbb{R}^{4\times 3}$ inside the ambient 4-dimensional space carrying a bilinear form q, and complete W to a basis with a scalar direction e_0 for S: $V = [\,S\mathrm{e}_0 \mid W\,]$.

e_0 is not a free choice. Whenever the 3×3 Gram $W^\mathsf{T} q W$ is non-degenerate (the same invertibility hypothesis Corollary 4.2 already requires), the q-orthogonal complement of $\mathrm{span}\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ is exactly one-dimensional, a standard fact of bilinear algebra: for non-degenerate q on the full 4-space, $\dim W + \dim W^{\perp_q} = 4$, and $W \cap W^{\perp_q} = \{0\}$ precisely because $q|_W$ is non-degenerate. We fix e_0 to be this unique direction (normalized, with the residual sign ambiguity the same harmless kind as P1's gauge). With this choice, $\Gamma_s := V^\mathsf{T} q V$ is automatically block-diagonal:

$$\Gamma_s = \begin{pmatrix} S^2\, q(\mathrm{e}_0, \mathrm{e}_0) & 0 \\ 0 & W^\mathsf{T} q W \end{pmatrix},$$

i.e. the congruence reading collapses, by construction, to the per-slot reading $\Gamma_s = \mathrm{diag}\big(q_S(S\mathrm{e}_0), q_A(\mathrm{A}), q_I(\mathrm{I}), q_R(\mathrm{R})\big)$ used in the companion instantiation work: the two are not independent alternatives licensed separately by A2; the per-slot reading is the congruence reading evaluated at the one gauge consistent with treating S as grade-0 (no cross-coupling to the vector slots, matching the grade-mismatch fact that S and a grade-1 attribute cannot pair under a grade-respecting product). Any other choice of e_0 gives the same signature by Sylvester (Corollary 4.2 below does not depend on which invertible V is used) but populates spurious S–A, S–I, S–R cross-entries with no counterpart anywhere they are actually used, so the orthogonal e_0 is not merely a valid gauge; it is the canonical one, singled out by consistency with every explicit construction in this program. Verification (existence, uniqueness, and the block-diagonal collapse, 5 random trials): models/calcs/brainstorming/papers/weld_clifford/puente_simbolo_gram_sylvester_prueba.py, part IV.

Axiom A3 (continuous evolution). The ODU evolves smoothly in time τ and space x, with a finite propagation speed $c > 0$. Treating $\Gamma(\tau, \mathrm{x})$ as a field and expanding to second order in both τ and x, consistent with A1 and A2, the generic equation of motion is

$$\ddot{\Gamma} + \gamma\dot{\Gamma} - c^2\nabla^2_{\mathrm{x}}\Gamma + \nabla_\Gamma P(\Gamma) = N(\Gamma),$$

where $\gamma \geq 0$ is a constitutive damping parameter, $\nabla^2_{\mathrm{x}} = \partial^2_{x_1} + \partial^2_{x_2} + \partial^2_{x_3}$ is the spatial Laplacian (propagation at speed c), and $\nabla_\Gamma P$ is the gradient of the structural potential in configuration space. This is the lowest-order equation coupling inertia ($\ddot{\Gamma}$), spatial propagation ($-c^2\nabla^2_{\mathrm{x}}\Gamma$), dissipation ($\gamma\dot{\Gamma}$), and restoring forces ($\nabla_\Gamma P$); no additional postulate about dynamics is made beyond smoothness and finite speed.

Remark 2.2. The genuine content of A2 is the claim that structure is the geometric product. A3 adds the claim that evolution is smooth and second-order: position and velocity are independent degrees of freedom, so a first-order equation would conflate them. The symmetric/antisymmetric split of the geometric product is a theorem of geometric algebra, not an additional hypothesis.

Remark 2.3 (the EOM does not retire Γ_s, Γ_a). A3's equation is written for the undivided Γ, and from here on the paper works mostly with Γ as a single matrix, which makes it easy to read this as A2's Force/Field split being used once, in §2, and then abandoned. It is not: every term of the EOM acts on both sectors simultaneously, because $\ddot{\Gamma} = \ddot{\Gamma}_s + \ddot{\Gamma}_a$, $\nabla^2_{\mathrm{x}}\Gamma = \nabla^2_{\mathrm{x}}\Gamma_s + \nabla^2_{\mathrm{x}}\Gamma_a$, and likewise for $\gamma\dot{\Gamma}$, by linearity of $\Gamma \mapsto \Gamma_s, \Gamma_a$. The one term that is not sector-blind is the potential: $P(\Gamma)$ in this paper's scope (§6, and Definition 2.1 of the companion atlas work) is a functional of Γ_s alone. The Force sector supplies the restoring force, and Γ_a is source-free and dissipation-free at this order, evolving only by inertia and propagation. So the decomposition is not lost; it reappears as a statement about which terms of the EOM each sector feels. This is used below without further comment (§6.1–6.2 recover Newton and

Maxwell as, respectively, the Γ_s-only and Γ_a-only limits of the same equation) and is made fully explicit, with the spectral consequence at $\det \Gamma_s = 0$, in the companion atlas work.

---

3. Four Lemmas

Lemma 1 (Closure: dimension is forced to $d = 3$)

Lemma 1. Under A1 and A2, the dimension of the vector attribute space is $d = 3$.

Proof (closure argument). A2 says the Field is the grade-2 part of the geometric product: $\mathcal{F} = \mathrm{I} \wedge \mathrm{R} \in \Lambda^2(\mathbb{R}^d)$, a bivector of dimension $\binom{d}{2} = \frac{d(d-1)}{2}$. For the ODU to be closed, for $\mathcal{F}$ to couple back to the grade-1 attribute A without introducing objects of higher rank than those in A1, the Field space and the attribute space must be isomorphic as vector spaces:

$$\binom{d}{2} = d \implies \frac{d(d-1)}{2} = d \implies d = 3.$$

The unique non-trivial solution is $d = 3$. This isomorphism is the Hodge duality $\star : \Lambda^2(\mathbb{R}^3) \xrightarrow{\sim} \mathbb{R}^3$, which maps $\mathcal{F} = \mathrm{I} \wedge \mathrm{R}$ to the familiar vector cross product $\mathrm{I} \times \mathrm{R} \in \mathbb{R}^3$.

Confirmation by Hurwitz. Once $d = 3$ is established by closure, the Eckmann theorem (Eckmann 1943; Adams 1960) confirms that a non-degenerate vector cross product on $\mathbb{R}^3$ exists: it is the imaginary part of the quaternion product ($\mathbb{H}$, dimension $d + 1 = 4$). This is a consistency check, not the source of the derivation: Hurwitz verifies that $d = 3$ works, but closure is what forces it.

The octonionic branch ($d = 7$) is structurally distinct. A cross product on $\mathbb{R}^7$ exists (Eckmann 1943) as the imaginary part of the octonion product, but it does not arise from the closure condition $\binom{d}{2} = d$ (since $\binom{7}{2} = 21 \neq 7$). It is not the Hodge dual of a bivector; it is a structurally different object. This branch is not pursued further in this paper. The remainder of this paper works $d = 3$. □

Corollary 1.1. "Why exactly three vector attributes" is not a free parametric choice; it is the answer to "what dimension allows the Field to couple back to the Agents without rank escalation."

Lemma 2 (Algebra closure)

Lemma 2. In $d = 3$, the three grade-1 attributes $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ generate the full geometric algebra $G(3)$ of dimension $8 = 2^3$.

Proof. Three linearly independent vectors in $\mathbb{R}^3$ generate $G(3)$ by definition: the basis elements are $\{1, e_1, e_2, e_3, e_1e_2, e_2e_3, e_3e_1, e$ (grade 0 through 3). S occupies grade 0; $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ occupy grade 1; the bivectors (grade 2) and the pseudoscalar (grade 3) are generated by their products. No fifth grade-1 generator is available in $G(3)$: the grade-1 subspace has dimension 3. □

Lemma 3 (Time as a fourth direction)

Lemma 3. Under A3, the smooth evolution of the ODU requires a temporal direction ∂_τ that is independent of the three spatial attribute directions of A1. Together they span a 4-dimensional vector space V^4.

Proof. By A3, the configuration $\Gamma(\tau, \mathrm{x})$ is smooth in time τ and space x. Smoothness implies the existence of partial derivatives ∂_τ and ∇ (the spatial gradient along the attribute directions). Lemma 2 establishes that the spatial attribute space is $\mathbb{R}^3$, spanned by $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$. The temporal direction ∂_τ is independent of these three generators for three reasons: (i) it cannot be expressed as a spatial combination of A, I, R, since it differentiates the temporal argument, not the attribute frame; (ii) S is grade-0 (scalar), not a grade-1 direction; (iii) promoting one spatial attribute to the temporal role would break the closure condition $\binom{d}{2} = d$ that forces $d = 3$ in Lemma 1. Therefore ∂_τ is a fourth, independent direction.

Together $\{\mathrm{A}, \mathrm{I}, \mathrm{R}, \partial_\tau\}$ span a 4-dimensional vector space V^4. The Clifford algebra $\mathrm{Cl}(V^4, q)$ has dimension $2^4 = 16$; its smallest faithful real matrix representation is 4×4 (Bott periodicity, once the signature q of V^4 is fixed by Lemma 4). The ODU has a time axis because it evolves smoothly, not because spacetime is postulated. □

Postulate A3′ (co-location: named explicitly, not derived). A3 writes $\Gamma(\tau, \mathrm{x})$ as a field over external coordinates x with a spatial Laplacian ∇^2_{x}, while A1/Lemma 1 give an internal attribute space span$\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$. Lemma 3's phrase "the spatial attribute directions of A1" silently identifies the two. This identification is not forced by A1–A3 as stated: an ODU could in principle carry internal grade-1 attributes without those attributes coinciding, as a vector space, with the coordinates over which it propagates. We name the identification a fourth postulate, since the derivation needs it and it was previously used without being declared:

> A3′. The grade-1 attribute directions $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ are realized as tangent vectors of the same physical space over which Γ propagates; i.e. the internal attribute space of A1 and the external coordinate space of A3 are one and the same $\mathbb{R}^3$.

This is the natural reading whenever the attributes are ordinary spatial vectors (velocity, angular momentum, a relational displacement), objects that already transform as $SO(3)$ vectors under rotations of the physical space the unit occupies, which is what "grade-1" was meant to capture in A1. It stops being automatic once an attribute is not literally a spatial vector (e.g. the per-slot norms of §2's Definition, where q_S, q_I, q_R need not come from the coordinate metric at all). A3′ is therefore load-bearing precisely at the joint named in Lemma 3 and revisited in §8.4: without it, Lemma 4 constrains only the coordinate space x, and says nothing about $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$. With it, the $(3, 1)$ conclusion of Lemma 4 transfers onto the attribute space itself, which is what Corollary 4.2 (below) requires to even be a meaningful question.

Lemma 4 (Lorentzian signature from the wave operator)

Lemma 4. The Lorentzian signature $(3, 1)$ is forced by the principal symbol of the EOM (A3), not postulated.

Proof. By A3, the EOM explicitly contains the spatial Laplacian $-c^2\nabla^2_{\mathrm{x}}\Gamma$. The principal part, the highest-derivative terms, which govern propagation, is therefore the wave operator $\Box\Gamma = \ddot{\Gamma} - c^2\nabla^2_{\mathrm{x}}\Gamma$. Isotropy of the Laplacian across the three spatial directions follows directly from P1: since all physical invariants of Γ are $SO(3)$-invariant under rotations of $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$, no spatial direction is preferred, and the spatial part of the principal symbol must be $c^2|\mathrm{k}|^2$ (scalar in k, not a directionally biased tensor). This rules out any anisotropic operator. (A parabolic operator such as the heat equation $\partial_\tau\Gamma = c^2\nabla^2_{\mathrm{x}}\Gamma$ is first-order in time and does not treat ∂_τ as an independent generator on equal footing with the spatial ones; it is excluded.) Taking the Fourier symbol ($\partial_\tau \to i\omega$, $\nabla \to i\mathrm{k}$):

$$\sigma(\Box) = -\omega^2 + c^2|\mathrm{k}|^2 = \eta^{\mu\nu}p_\mu p_\nu, \quad \eta = \mathrm{diag}(-1, +1, +1, +1)$$

This is the Minkowski quadratic form with signature $(3, 1)$. The real Clifford algebra of this form (Atiyah, Bott and Shapiro 1964; Lounesto 2001) is $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$.

Uniqueness of the real representation. The Bott periodicity classification gives $\mathrm{Cl}_{3,1}$ as a real matrix algebra. Requiring Γ to be a real matrix (the gradient flow $\dot{\Gamma} = -\nabla P$ is real; dissipation is a real process) pins the signature to $(3, 1)$ and excludes $(1, 3)$ (which gives $\mathrm{Cl}_{1,3} \cong M_2(\mathbb{H})$, quaternionic), $(4, 0)$ (which gives $\mathrm{Cl}_{4,0} \cong M_2(\mathbb{H})$, also quaternionic), and all other real signatures that fail to give $M_4(\mathbb{R})$. The only signature that yields a real 4×4 matrix algebra with three spatial and one temporal generator is $(3, 1)$. □

Remark 3.1. The step "$d = 3$ forces real 4×4" is the logical bridge: asking for three spatial generators in a real matrix algebra forces exactly $M_4(\mathbb{R}) = \mathrm{Cl}_{3,1}$. The Lorentzian signature is not a choice; it is what remains after demanding reality and three spatial dimensions.

Remark 3.2 (The ladder $\mathrm{Cl}_{3,0} \to \mathrm{Cl}_{3,1} \to \mathrm{Cl}_{4,1}$). Lemma 3 is the general mechanism by which a UoC's host algebra grows: a new generator is added exactly when A3's smoothness condition exposes an independent direction that the existing generators cannot express. Newton's second law (§6.1) is the $d = 3$, no-time-generator floor: it needs no ∂_τ beyond an ordinary scalar time parameter, so its host is $\mathrm{Cl}_{3,0}$. Promoting ∂_τ to an independent grade-1 generator, forced once the EOM couples space and time symmetrically through the wave operator $\Box$ (Lemma 4), is exactly the step that produces $\mathrm{Cl}_{3,1}$, the host of free electrodynamics (§6.2). A further UoC, explored outside this paper under the name $\mathrm{UoC}_{\mathrm{st}}$ (spacetime as a dynamical unit in its own right, with ρ as a genuine fifth attribute rather than a derived quantity), exposes a second independent direction beyond the four of $\mathrm{Cl}_{3,1}$: a conformal/scale direction, verified numerically to require a spacelike generator ($e_\rho^2 = +1$) to keep the Gram signature Lorentzian, $(3, 1)$, rather than degrading to $(2, 2)$ under a timelike choice ($\mathrm{Cl}_{3,2}$). This fixes the host to $\mathrm{Cl}_{4,1}$, which contains $\mathrm{Cl}_{3,1}$ as the even subalgebra of its Clifford grading (§30.3 of the companion exploration). The pattern across all three steps is the same: the temporal/scale component is never postulated, each new generator is forced by a smoothness or signature-consistency condition applied to the previous algebra. The full numerical verification of the $\mathrm{Cl}_{4,1}$ vs. $\mathrm{Cl}_{3,2}$ signature comparison is given in the companion code repository (§7); this extension is not part of the closed theorem of this paper and is flagged as such in §8.4.

---

4. Main Theorem

Theorem (Γ is forced). Given axioms A1, A2, and A3, the configuration of any operative dynamical unit is necessarily

$$\boxed{\Gamma = \Gamma_s \oplus \Gamma_a \ \in \ M_4(\mathbb{R}) = \mathrm{Cl}_{3,1}}$$

where $\Gamma_s = \frac{1}{2}(\Gamma + \Gamma^\top)$ (symmetric, 10 independent entries; the Force sector) and $\Gamma_a = \frac{1}{2}(\Gamma - \Gamma^\top)$ (antisymmetric, 6 independent entries; the Field sector), with: - $\Gamma_s = \mathrm{Gram}(\mathrm{A}, \mathrm{I}, \mathrm{R}; S)$: the symmetric metric coupling structure of the attributes (Force sector; S is the grade-0 identity, not a grade-1 generator) - Γ_a = magnetic part ($\mathrm{I} \wedge \mathrm{R}$, spatial bivector, from SAIR) $\oplus$ electric part ($\partial_\tau \wedge \nabla$, spacetime bivector, coupling structure$\leftrightarrow$evolution)

Proof. Lemma 1 forces $d = 3$ by closure. Lemma 2 establishes $G(3)$. Lemma 3 adds the temporal generator ∂_τ, extending $G(3)$ to a 4-generator algebra. Lemma 4 identifies the resulting algebra as $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$ via the wave operator symbol.

Uniqueness of $\Gamma = \Gamma_s \oplus \Gamma_a$. A positive-definite symmetric matrix (metric) carries only Force ($\Gamma_s \succ 0$, no antisymmetric part). A symplectic form carries only Field (Γ_a, no symmetric part). The object $\Gamma = \Gamma_s \oplus \Gamma_a$ is the unique minimal object that carries both Force and Field simultaneously: the unique decomposition of a real 4×4 matrix into symmetric and antisymmetric parts. $\square$

Remark 4.1. The theorem does not claim that $\mathrm{Cl}_{3,1}$ is the algebra of spacetime as a physical background. It claims that any operative dynamical unit that satisfies A1 and A2 has a configuration object that is an element of $\mathrm{Cl}_{3,1}$. Physical spacetime is not an input; it is a limit (the space-time branch of the framework; see §6.2).

---

5. Three Closing Propositions

Three technical choices appear in the proof: the orthonormality of the attribute frame, the identification of the temporal generator, and the metric on $M_4(\mathbb{R})$. Each is forced, not free.

Proposition P1 (Orthonormality is gauge-redundant)

Proposition P1. The physical invariants of Γ, its determinant, trace, eigenvalues, and singular values, are independent of the choice of orthonormal frame for $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$. Orthonormality is a representational gauge, not a physical postulate.

Proof. Let $M \in SO(3)$ be the Gram-Schmidt rotation mapping any linearly independent triple $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ to an orthonormal frame $\{\hat{\mathrm{A}}, \hat{\mathrm{I}}, \hat{\mathrm{R}}\}$ spanning the same subspace. Under this rotation, $\Gamma \mapsto M\Gamma M^\top$. The spectrum of Γ (equivalently, $\det\Gamma$, $\mathrm{tr}\,\Gamma$, and all eigenvalues) is invariant under conjugation by any $M \in SO(3)$. The orthonormal frame fixes the explicit matrix entries of Γ_s (diagonal in that frame) but not its spectrum. $\square$

Proposition P2 (The temporal generator is the evolution operator)

Proposition P2. The temporal Clifford generator $\gamma_0 \in \mathrm{Cl}_{3,1}$ satisfying $\gamma_0^2 = -1$ is the unique grade-1 generator assigned to the temporal direction such that the linear differential operator $\mathcal{D} = \gamma^\mu \partial_\mu$ factorizes the wave operator: $\mathcal{D}^2 = \square$.

Proof. The symbol of the wave operator $\square = \partial_\tau^2 - c^2\nabla^2$ is $\eta^{\mu\nu} p_\mu p_\nu$ with $\eta = \mathrm{diag}(-1, +1, +1, +1)$. This defines the anticommutation relations $\{\gamma_\mu, \gamma_\nu\} = 2\eta_{\mu\nu}$, with $\gamma_0^2 = -1$ forced by $\eta_{00} = -1$. The assignment of γ_0 to the temporal direction ∂_τ is the unique one that makes $(\gamma^\mu \partial_\mu)^2 = \square$ (the Dirac factorization). Here γ_0 is an algebra element and ∂_τ is a differential operator; P2 does not claim they are the same object, only that γ_0 is the algebraic generator conjugate to ∂_τ in the factorization of $\square$. Verified numerically: a real 4×4 representation satisfying $\{\gamma_\mu, \gamma_\nu\}/2 = \eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ exists with $\gamma_0^2 = -I$, $\gamma_i^2 = +I$ (residual $< 10^{-14}$; see code/verify_cl31.py). $\square$

Note. P2 does not conflate categories: γ_0 is an element of $\mathrm{Cl}_{3,1}$; ∂_τ is a differential operator acting on functions $\Gamma(\tau, \mathrm{x})$. What P2 establishes is a canonical pairing between the two, mediated by the Dirac factorization of the wave operator given by A3.

Proposition P3 (Frobenius is the canonical Clifford metric)

Proposition P3. The metric on the space of configurations $\Gamma \in M_4(\mathbb{R}) = \mathrm{Cl}_{3,1}$ is the Frobenius norm $\|\Gamma\|^2 = \mathrm{Tr}(\Gamma^\top \Gamma)$, already determined by A2.

Proof. The geometric product of A2 defines the canonical inner product on any Clifford algebra:

$$\langle A, B \rangle_{\mathrm{Cl}} := \langle A\tilde{B} \rangle_0$$

the grade-0 (scalar) component of the geometric product of A with the Clifford reverse $\tilde{B}$. In the real 4×4 representation of $\mathrm{Cl}_{3,1}$, the Clifford reverse corresponds to matrix transposition ($\tilde{B} = B^\top$), giving $\langle A, B \rangle_{\mathrm{Cl}} = \frac{1}{4}\mathrm{Tr}(A^\top B)$, which is the Frobenius inner product up to the normalization $\frac{1}{4}$.

Uniqueness. $\mathrm{Spin}(3, 1)$ acts on $\mathrm{Cl}_{3,1}$ by adjoint conjugation $g \cdot \Gamma = g\Gamma g^{-1}$. Any physical metric must be invariant under this action. The grade decomposition $\Lambda^0 \oplus \Lambda^1 \oplus \Lambda^2 \oplus \Lambda^3 \oplus \Lambda^4$ of $\mathrm{Cl}_{3,1}$ consists of pairwise non-isomorphic irreducible representations of $\mathrm{Spin}(3, 1)$. By Schur's lemma, any $\mathrm{Spin}(3, 1)$-invariant bilinear form is proportional to $\mathrm{Tr}(A^\top B)$ on each grade-block, with a possibly different constant on each grade. The submultiplicativity constraint, $\|\Gamma\Gamma'\| \leq \|\Gamma\|\|\Gamma'\|$ for all $\Gamma, \Gamma' \in \mathrm{Cl}_{3,1}$, required for $P(\Gamma) = \|\Gamma\|^2$ to be compatible with the algebra product, forces equal scaling across all grades (a different scale per grade would violate submultiplicativity for mixed-grade products). The result is Frobenius, uniquely. □

Extension to $\mathrm{Cl}_{4,1}$. The spacetime configuration in the GSF framework lives in $\mathrm{Cl}_{4,1} \cong M_4(\mathbb{C})$ (Bott periodicity: $p - q = 3$). The same argument applies with the Clifford reverse replaced by the Hermitian conjugate: $\langle A, B \rangle = \frac{1}{4}\mathrm{Tr}(A^\dagger B)$, giving the Hilbert-Schmidt norm $\|\Gamma\|^2 = \mathrm{Tr}(\Gamma^\dagger \Gamma)$, real and non-negative. P3 holds without modification.

---

6. Two Physical Limits

The main theorem and propositions establish the algebraic structure. Two physical theories appear as limiting cases, showing the algebra has concrete content.

6.1 Newton's second law (Force sector, det > 0)

In the operationally active sector ($\det\Gamma > 0$, $\Gamma_s \succ 0$), the dominant dynamics is along the soft mode of Γ, the direction of smallest singular value. Project the EOM onto the soft-mode scalar x (dominant singular value of Γ along A):

$$\ddot{x} + \gamma\dot{x} + \partial_x P = F_{\mathrm{ext}}$$

This is Newton's second law for a damped oscillator: mass is the inertia of the soft mode, γ is the damping of the ODU, and P is the potential landscape. Newton's law is not a special case built into the framework; it is what the EOM becomes when projected onto the dominant mode. Status: structural correspondence ⟨CE⟩, not a derivation from first principles.

6.2 Free electrodynamics (Field sector, det = 0 boundary)

At the boundary $\det\Gamma = 0$, the configuration loses rank: one or more singular values of Γ approach zero. When $\gamma \to 0$ (non-dissipative limit) and the configuration lives in the grade-2 (bivector) sector Γ_a of $\mathrm{Cl}_{3,1}$, the EOM reduces to:

$$\Box\Gamma_a = 0 \quad \Rightarrow \quad \Box F_{\mu\nu} = 0$$

which is the free Maxwell equation in the absence of sources ($\partial^\mu F_{\mu\nu} = 0$), together with the Bianchi identity $\partial_{[\mu}F_{\nu\rho]} = 0$ (automatic from $F = \mathrm{d}A$). The bivector $F_{\mu\nu}$ is the Faraday tensor; the identification of Γ_a with the grade-2 sector of $\mathrm{Cl}_{3,1}$ makes the Field $\mathrm{I} \wedge \mathrm{R}$ correspond to the spatial (magnetic) part of $F_{\mu\nu}$, and the electric part arises from the $\partial_\tau \wedge \nabla$ coupling. In the free-field sector, charge conservation $\partial^\mu J_\mu = 0$ follows from the antisymmetry of F without additional assumptions. Status: free Maxwell ⟨TEO⟩[D]; source equation ⟨A⟩ (requires closing the cross-coupling block).

Remark 6.1 (reconciling two decompositions of F). I, R here are ordinary grade-1 generators of V^4 (Lemma 3), the same status as A, I, R in A1, and $\mathrm{I} \wedge \mathrm{R}$ is a genuine wedge of two grade-1 vectors, exactly the mechanism that produces $\mathrm{L} = \mathrm{I} \wedge \mathrm{R}$ in the Newton limit (§6.1) or vorticity in the Navier–Stokes correspondence. B is grade-2 already in $\mathbb{R}^3$ (it is dual to the ordinary grade-1 magnetic vector, i.e. an axial vector), so identifying it with $\mathrm{I} \wedge \mathrm{R}$ requires no change to A1. E, by contrast, is grade-1 in $\mathbb{R}^3$; it only becomes a bivector component once lifted to V^4 by wedging with the temporal generator ($\partial_\tau \wedge \nabla$, not $\mathrm{I} \wedge \mathrm{R}$). A companion exploration (Pieza 4, pieza4_electromagnetismo.md) labels the two covariant halves of the already-assembled Faraday bivector F as "$I = E$, $R = B$" under a work-based criterion (which component does work on a charge). That labeling answers a different question, how to split F once it exists, and is not in tension with the derivation here, which answers how F's magnetic half is built from two grade-1 generators. The two readings share names but not referents; they should not be conflated.

---

7. Numerical Verification

The following steps in the derivation are numerically confirmed; scripts are in code/:

Result	Script	Residual
$\{\gamma_\mu, \gamma_\nu\}/2 = \eta_{\mu\nu}$ in real 4×4 rep	verify_cl31.py	$< 10^{-14}$
$\gamma_0^2 = -I, \gamma_i^2 = +I$	verify_cl31.py	$< 10^{-14}$
$\langle A, B\rangle_{\mathrm{Cl}} = \mathrm{Tr}(A^\top B)/4$ on grade-1 elements	verify_clifford_metric.py	$< 10^{-14}$
Frobenius submultiplicativity: $\|\Gamma\Gamma'\|_F \le \|\Gamma\|_F \|\Gamma'\|_F$ (0 violations); Pythagorean: $\|\Gamma\|^2 = \|\Gamma_s\|^2 + \|\Gamma_a\|^2$	verify_frobenius.py	0 violations; error $< 10^{-13}$
$\det\Gamma$ as invariant under $SO(3,1)$ conjugation	verify_det_invariance.py	$< 10^{-12}$

---

8. Discussion

8.1 What is new

The spacetime algebra program (Hestenes 1966; Doran and Lasenby 2003) takes $\mathrm{Cl}_{3,0}$ or $\mathrm{Cl}_{1,3}$ as the algebra of physical space or spacetime, motivated by the known geometry. The present work reverses this logic: $\mathrm{Cl}_{3,1}$ is derived as the forced algebraic structure of any self-describing dynamical unit, without assuming a spacetime background. The key steps are: (i) the closure condition $\binom{d}{2} = d$ forces the dimension of the vector attribute space (Hurwitz confirms consistency); (ii) the wave operator fixes the signature; (iii) the reality of Γ selects $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$ over the vector-space-isomorphic but algebra-distinct $\mathrm{Cl}_{1,3} \cong M_2(\mathbb{H})$. Step (iii) is a selection between signature conventions that a metric-first derivation does not face and that other emergent-signature routes (§8.3) do not make; it is specific to the demand that the configuration and its dissipative dynamics be real. Neither step is obvious from the physics-first perspective.

The closest algebraic antecedent is the observation (Lounesto 2001, §17) that the Clifford algebra of the symbol of the wave operator is $\mathrm{Cl}_{3,1}$. Here that observation becomes a theorem: within axioms A1–A3 it is the only Clifford algebra consistent with a real, smoothly-evolving self-describing unit.

8.2 Corollary: the signature result exhausts the alternatives

Lemma 4 selects $(3, 1)$. The same argument, run over the full space of candidate symbols rather than only the one A3 produces, classifies the alternatives rather than merely excluding them, which strengthens the theorem at no additional cost.

Corollary 4.1 (completeness of the regimes). Let q be the principal symbol of a second-order EOM on V^4, i.e. a real quadratic form on a 4-dimensional space. Then: (i) by Sylvester's law of inertia, q falls into exactly one of 15 congruence classes indexed by (n_+, n_0, n_-) with $n_+ + n_0 + n_- = 4$; the non-degenerate ones form 5 connected components of the space of such forms, since the inertia is a complete and locally constant invariant; (ii) modulo the global sign convention $(n_+, \cdot, n_-) \sim (n_-, \cdot, n_+)$, exactly three regimes remain: elliptic $(4,0)$, hyperbolic $(3,1)$, ultrahyperbolic $(2,2)$; (iii) of these, exactly one, the Lorentzian $(3,1)$, yields a well-posed Cauchy problem. The elliptic case is well-posed as a boundary-value problem but describes equilibrium, not evolution; the ultrahyperbolic case has two temporal directions and is Hadamard ill-posed.

Proof. (i) is Sylvester's law together with continuity of eigenvalues: a path between forms of different inertia must pass through a degenerate form, so each class is open and closed in the non-degenerate stratum. (ii) is the identification of a form with its negative. (iii) is the standard PDE classification by principal symbol (Courant and Hilbert 1962, vol. II): all eigenvalues of one sign gives an elliptic operator; exactly one of opposite sign gives a hyperbolic operator with well-posed Cauchy data; two or more of each gives an ultrahyperbolic operator, for which the Cauchy problem is ill-posed. □

Two consequences follow for the reading of Lemma 4. First, the uniqueness of $(3,1)$ is not a statement about a short list of physically motivated candidates: the list of all real signatures on V^4 is finite, is exhausted above, and $(3,1)$ is the only survivor. Second, the excluded cases acquire meaning rather than merely being ruled out: the elliptic class is the static/equilibrium regime, and the ultrahyperbolic class is the genuine pathology. This matters because det-based classifications cannot see the distinction: $\det q > 0$ holds for both $(4,0)$ and $(2,2)$, so the determinant sign merges a physical regime with a pathological one, and only the full inertia separates them. A companion paper uses precisely this stratification to organise the dynamical regimes of Γ.

Scope. Corollary 4.1 concerns the principal symbol, the object that fixes PDE type and well-posedness. It should not be conflated with the inertia of the Gram matrix Γ_s of A2, which is a different object built from the attribute slots. The two signatures are, in general, independent invariants of different objects; Corollary 4.2 states exactly when they must agree.

Corollary 4.2 (when the Gram inherits the symbol's signature). Let η be the symbol's $(3,1)$ form (Lemma 4, under A3′) and let Γ_s be built by the congruence reading of the Definition in §2 (Axiom A2), $\Gamma_s = V^{\mathsf{T}}\eta V$ with $V = [S\mathrm{e}_0 \mid \mathrm{A} \mid \mathrm{I} \mid \mathrm{R}]$. Then $\mathrm{signature}(\Gamma_s) = \mathrm{signature}(\eta) = (3,1)$ whenever V is invertible, with no further condition.

Proof. This is exactly Sylvester's law of inertia: congruence by an invertible matrix preserves signature. □

The condition is sharp. With e_0 fixed by the gauge closure of §2, the per-slot reading with heterogeneous forms $q_S, q_I, q_R \neq \eta$, used whenever the four attributes are not all measured under the one form η (e.g. $S, \mathrm{I}, \mathrm{R}$ carrying ordinary positive-definite norms with no Minkowski content), is not a congruence of η at all (it is still a congruence of some block form, namely $q_S \oplus q_A \oplus q_I \oplus q_R$, but not of the single η Corollary 4.2 requires). Corollary 4.2 does not apply there, and Γ_s's signature is free to range over all five classes of Corollary 4.1(i); this is confirmed numerically in the companion instantiation work, where a massive (particle) state lands on $(4,0)$, a photon state on the degenerate boundary, and a general-relativistic state, where the attribute A genuinely is a Minkowski four-velocity, satisfying $\langle \mathrm{A}, \mathrm{A}\rangle_\eta = -c^2$, on $(3,1)$, the one case closest to satisfying Corollary 4.2's hypothesis. Verification: models/calcs/brainstorming/papers/weld_clifford/puente_simbolo_gram_sylvester_prueba.py.

Remark (no residual mystery). The apparent tension, "the symbol forces $(3,1)$" versus "the Gram ranges over five sectors," dissolves once the two constructions of §2 are told apart. There is no hidden inconsistency: the symbol's $(3,1)$ is a fixed background fact about the operator (governs whether evolution of the field $\Gamma(\tau, \mathrm{x})$ is well-posed); the Gram's signature is a state-dependent fact about the value Γ_s takes at an instant (classifies the regime of that state). They coincide, by theorem, exactly on the congruence reading with V invertible; they are free to differ, also by theorem (Sylvester simply does not constrain a non-congruence construction), under the per-slot reading. Which reading a given physical domain uses is a modelling fact about that domain, not a gap in the algebra.

8.3 Relation to emergent-signature approaches

The idea that the Lorentzian signature should be derived rather than postulated is not new, and this paper does not claim priority for that program; it contributes a specific route. Two comparisons fix the position of the present derivation.

Singh (2025) obtains a Lorentzian signature within an octonionic pre-spacetime theory by adopting split division algebras: the split-complex unit ω with $\omega^2 = +1$ gives a magnitude $x^2 - y^2$ (Lorentzian) in place of $x^2 + y^2$ (Euclidean), and split bioctonions then generate a base of signature $(3,3)$ carrying embedded 4-dimensional Lorentzian spacetimes. The signature there is a consequence of choosing split algebras, a choice motivated by the target signature. The present derivation differs in two respects. First, the mechanism: the signature is fixed by the requirement that the second-order equation of motion (A3) admit a well-posed Cauchy problem, a Lorentzian principal symbol is the only one that does (Lemma 4; Hadamard), rather than by selecting a split number system. No split-ness is assumed; the sign flip is forced by hyperbolicity of the evolution. Second, the target algebra is pinned to $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$ specifically, not the isomorphic-as-vector-space but distinct-as-algebra $\mathrm{Cl}_{1,3} \cong M_2(\mathbb{H})$, by the reality of Γ (dissipative and gradient dynamics are real processes; §8.1, §4). Singh's construction lives in higher dimension $(3,3)$ with embedded Lorentzian slices and does not make this reality-of-configuration selection between the two signature conventions. The two derivations are therefore complementary: both answer "yes" to the theorem-vs-postulate question, by independent mechanisms, and the present one rests on the more economical premise: well-posedness of a real evolution, rather than a chosen split algebra.

More broadly, the view of an emergent Lorentzian signature has a long tradition in analogue and induced gravity (Sakharov 1967; Barceló, Liberati and Visser 2011; Volovik 2003), where the effective Lorentzian metric arises from the low-energy behaviour of a non-relativistic substrate. The present result is narrower and purely algebraic: it does not construct an effective metric from a substrate, but identifies which real Clifford algebra the structure of a self-describing dynamical unit forces. It is offered as a structural companion to those programs, not a replacement.

8.4 Honest scope

This paper establishes the algebraic structure. It does not: - Prove uniqueness of the SAIR attribute structure independent of A1 (that is the content of A1 itself, which we take as foundational rather than derived) - Close the $d = 7$ (octonionic) branch (open problem; not pursued in this paper) - Close the further ladder step $\mathrm{Cl}_{3,1} \to \mathrm{Cl}_{4,1}$ sketched in Remark 3.2: the conformal/scale generator's necessity is verified numerically in the companion exploration but not yet derived from A1–A3 with the same rigor as Lemmas 1–4 - Establish instance-level uniqueness of the SAIR slot assignment (Remark 2.1's qualification): which observable quantity of a given ODU fills S versus $\mathrm{A}, \mathrm{I}, \mathrm{R}$ is not decided by the container theorem. A necessary condition (representation compatibility, Schur) follows from the same argument as Lemmas 2–4; sufficient selection criteria are the subject of a companion, less mature line of work (seven domains checked by blind retrodiction and active rejection of incorrect-but-compatible assignments) that is intentionally kept out of this paper rather than diluting its closed lemmas with an open one - Establish, for a given domain, existence of a SAIR quadruple in the first place (Remark 2.1's second qualification): whether $\mathrm{A}, \mathrm{I}, \mathrm{R}$ arise as grade-1 vectors at all, with Force and Field genuinely inherent, is empirical, not algebraic, and is not established here for any domain. This is logically prior to, and independent of, the uniqueness question above; the container theorem is conditional on existence and proves nothing about it

Two structural gaps in the axiom set are addressed directly below, each with a precise resolution rather than a bare admission.

(a) The attribute space and the coordinate space, identified without argument: Postulate A3′. A1 and Lemma 1 give an internal attribute space $\mathbb{R}^3$ spanned by $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$. A3 treats $\Gamma(\tau, \mathrm{x})$ as a field over an external coordinate space carrying ∇^2_{x}, and Lemma 4 reads the signature off that Laplacian. The step identifying the two spaces is named explicitly, immediately after Lemma 3, as Postulate A3′. The identification is not derivable from A1–A3 as stated (an ODU could in principle carry internal attributes that do not coincide with its propagation coordinates), so it is stated as a fourth, independent premise, on the same footing as A1's own irreducibility (Remark 2.1). The theorem's reach was always conditional on this identification; naming it as A3′ lets the reader see the condition and evaluate it, rather than find it absorbed silently inside "the spatial attribute directions of A1" in Lemma 3.

(b) Two distinct objects, both called "the signature": Corollary 4.2. Lemma 4 and Corollary 4.1 concern the inertia of the principal symbol on $V^4 = \mathrm{span}\{\mathrm{A}, \mathrm{I}, \mathrm{R}, \partial_\tau\}$. A2 independently supplies a Gram matrix Γ_s of the slots $\{S, \mathrm{A}, \mathrm{I}, \mathrm{R}\}$ (§2, Definition), which also carries an inertia. These need not agree in general: they are different 4-spaces, and the Gram's construction admits two readings (§2). Corollary 4.2 states the exact condition under which they do agree: the Gram inherits the symbol's $(3, 1)$ signature, by Sylvester's law of inertia, precisely when it is built as a congruence $V^\top \eta V$ with V invertible. Under the alternative per-slot construction, also licensed by A2, and the one used whenever an attribute carries no Minkowski structure of its own, no such inheritance holds, and the signature ranges over all admissible classes, matching what a companion paper observes when it reads dynamical regimes off the Gram. The two readings of the Definition in §2 were simply not told apart before.

The residues that remain open after P1/P2/P3 are the three original axioms A1, A2, A3, together with the newly named Postulate A3′. Gap (a) is resolved by declaring A3′ as a premise, whose truth for a given physical domain is a modelling question, not a theorem. Gap (b) is resolved outright: Corollary 4.2 is a theorem with a checkable hypothesis, not an open problem. The axiom count grows from three to four, but no premise enters the derivation undeclared, and the two notions of "signature" can no longer be silently conflated.

8.5 Related work

- Hurwitz (1898): normed division algebras in dimensions 1, 2, 4, 8.
- Eckmann (1943): cross products in $\mathbb{R}^n$ exist only for $n = 1, 3, 7$.
- Atiyah, Bott and Shapiro (1964): Clifford modules and Bott periodicity.
- Hestenes (1966, 1986): spacetime algebra as the language of physics.
- Doran and Lasenby (2003): geometric algebra for physicists (Cambridge).
- Lounesto (2001): Clifford algebras and spinors (Cambridge).
- Adams (1960): vector fields on spheres; confirms Hurwitz via K-theory.
- Sakharov (1967): induced gravity; the metric as an emergent elastic response.
- Barceló, Liberati and Visser (2011): analogue gravity (Living Rev. Relativity); emergent Lorentzian metrics from non-relativistic substrates.
- Volovik (2003): The Universe in a Helium Droplet (Oxford); emergent relativity and effective metric in quantum liquids.
- Singh (2025): trace dynamics, octonions and unification; Lorentzian signature from split bioctonions. arXiv:2501.18139.
- Molina (2024a): the determinant as the source of the cubic term in matrix gradient flows. DOI: 10.5281/zenodo.20752208

---

9. Conclusions

From four structural axioms (SAIR attribute structure (A1), geometric product (A2), continuous smooth evolution (A3), and co-location of the attribute space with the propagation coordinates (A3′)), the real Clifford algebra $\mathrm{Cl}_{3,1}$ emerges as a structural theorem rather than a geometric postulate. The derivation chain is:

$$\underbrace{\text{SAIR}}_{\text{A1}} + \underbrace{\text{geom. product}}_{\text{A2}} \xrightarrow{\binom{d}{2}=d} d = 3 \xrightarrow{G(3)} \{\mathrm{A}, \mathrm{I}, \mathrm{R}\} = \gamma_i \xrightarrow{\text{A3: smooth, finite } c} \partial_\tau \perp \mathbb{R}^3 \xrightarrow{\sigma(\Box)=\eta} (3,1) \xrightarrow{\text{P1/P2/P3}} \Gamma = \Gamma_s \oplus \Gamma_a \in M_4(\mathbb{R})$$

Three propositions close the technical residues: orthonormality is a representational gauge (P1); γ_0 is the unique algebraic generator conjugate to ∂_τ in the Dirac factorization of $\Box$, not the operator itself (P2); Frobenius is the canonical Clifford metric forced by A2 (P3). The irreducible axioms are A1, A2, and A3. Classical mechanics (§6.1) and free electrodynamics (§6.2) appear as structural limits.

The result provides a structural basis for the spacetime algebra program: not "given Minkowski space, use $\mathrm{Cl}_{3,1}$", but "from the structure of a dynamical unit, $\mathrm{Cl}_{3,1}$ is the forced representation." This last step is conditional on the unit having a SAIR quadruple to begin with (§2, §8.4); the theorem forces the algebra, not the existence of the attributes it acts on.

References

Adams, J. F. (1960). Vector fields on spheres. Annals of Mathematics, 75(3), 603–632.

Atiyah, M. F., Bott, R., and Shapiro, A. (1964). Clifford modules. Topology, 3(S1), 3–38.

Doran, C. and Lasenby, A. (2003). Geometric Algebra for Physicists. Cambridge University Press.

Eckmann, B. (1943). Stetige Lösungen linearer Gleichungssysteme. Commentarii Mathematici Helvetici, 15(1), 318–339.

Hestenes, D. (1966). Space-Time Algebra. Gordon and Breach.

Hestenes, D. (1986). A unified language for mathematics and physics. In Clifford Algebras and their Applications in Mathematical Physics. D. Reidel.

Hurwitz, A. (1898). Über die Komposition der quadratischen Formen von beliebig vielen Variabeln. Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, 309–316.

Lounesto, P. (2001). Clifford Algebras and Spinors (2nd ed.). Cambridge University Press.

Molina, H. (2024a). The determinant as an orientation invariant and the source of the cubic term in equivariant matrix gradient flows. DOI: 10.5281/zenodo.20752208

Molina, H. (2025). Gamma Space Framework (working manuscript). Available at: https://github.com/hmolinab/gamma-space-framework